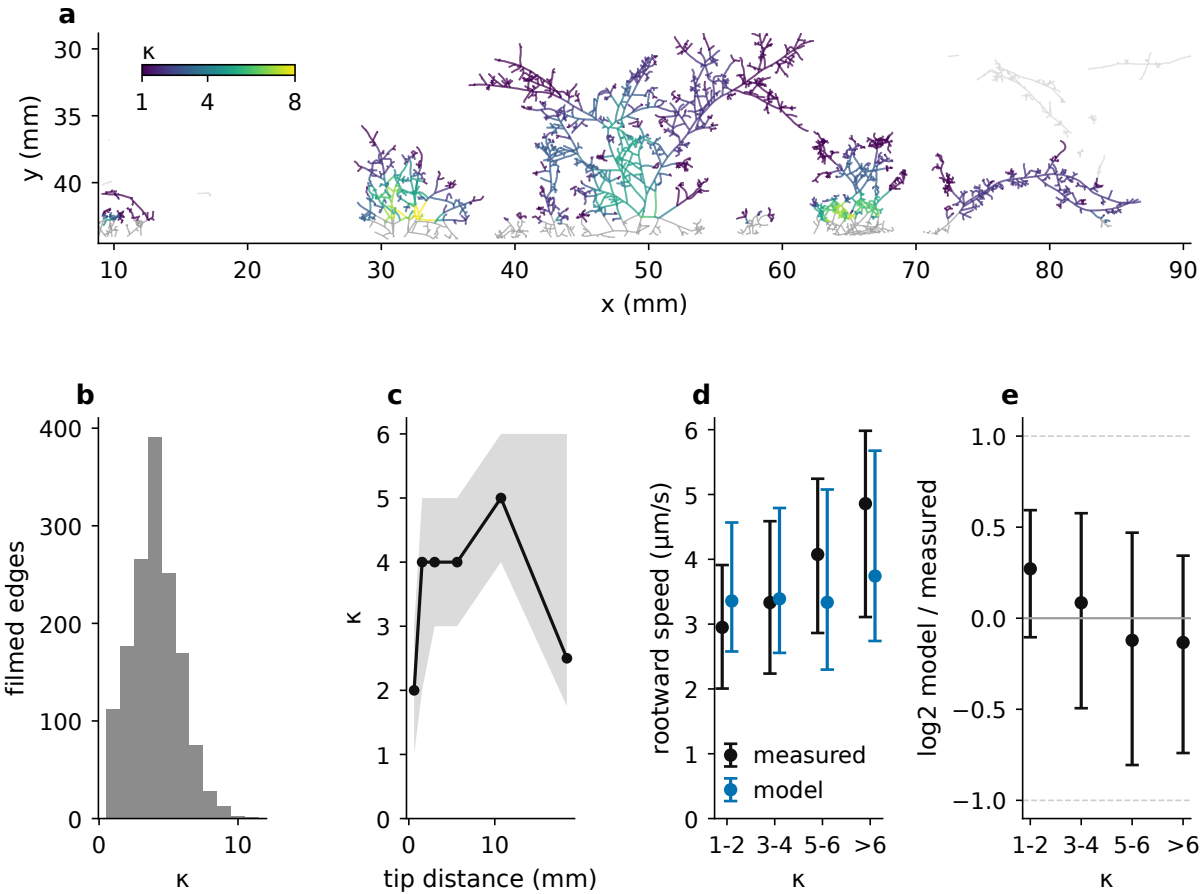

a
κ
1
4
8
y (mm)
30
35
40
10
20
30
40
50
60
70
80
90
x (mm)
b
filmed edges
400
300
200
100
0
0
10
κ
c
κ
6
5
4
3
2
1
0
0
10
tip distance (mm)
d
rootward speed (µm/s)
6
5
4
3
2
1
0
measured
model
1-2
3-4
5-6
>6
κ
e
log2 model / measured
1.0
0.5
0.0
−0.5
−1.0
1-2
3-4
5-6
>6
κ